

RESEARCH ARTICLE

Integrated application of metabolomics and RNA-seq reveals thermogenic regulation in goat brown adipose tissues

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Abstract

Brown adipose tissue (BAT) plays an important role on non-shivering thermogenesis during cold exposure to maintain animal body temperature and energy homeostasis. However, knowledge of the cellular transition from white adipose tissue (WAT) to BAT is still limited. In this study, we provided a comprehensive metabolomics and transcriptional signatures of goat BAT and WAT. A total of 157 metabolites were significantly changed, including 81 upregulated and 76 downregulated metabolites. In addition, we identified the citric acid cycle, fatty acid elongation, and degradation pathways as coordinately activated in BAT. Interestingly, five unsaturated fatty acids (Eicosadienoic Acid, C20:2; γ -Linolenic acid, C20:3; Arachidonic Acid, C20:4; Adrenic acid, C22:4; Docosahexaenoic acid, C22:6), Succinate, L-carnitine, and L-palmitoyl-carnitine were found to be abundant in BAT. Furthermore, L-carnitine, an intermediate of fatty acid degradation, is required for goat brown adipocyte differentiation and thermogenesis through activating AMPK pathway. However, L-carnitine decreased lipid accumulation through inducing lipolysis and thermogenesis in white adipocytes. These results revealed that there are the significant alterations in transcriptomic and metabolomic profiles between goat WAT and BAT, which may contribute to better understanding the roles of metabolites in BAT thermogenesis process.

KEYWORDS

brown adipose tissue, goat, L-carnitine, metabolomics, RNA-seq, unsaturated fatty acid

1 | INTRODUCTION

In mammals, three types of adipose tissue have recently been discovered, namely white adipose tissue (WAT), beige adipose tissue (Brite) and brown adipose tissue

(BAT).¹ WAT is single chamber and stored with large lipid droplets, while BAT is enriched with mitochondria, and its structural features are multi-chambered and small lipid droplets.² Additionally, when WAT in response to cold exposure and other stimulating conditions such as

Abbreviations: BAT, brown adipose tissue; Brite, beige adipose tissue; FDR, false discovery rate; OPLS-DA, orthogonal partial least-squares-discriminant analysis; PCA, principal component analysis; TCA, the citric acid cycle; UCP1, uncoupling protein 1; WAT, white adipose tissue.

exercise, hormones and so on, Brite has been found in WAT and the uncoupling protein 1 (UCP1) has a higher expression in Brite compared to WAT, which promotes energy consumption and exerts the role of heat production.^{3,4} WAT is mainly responsible for metabolic energy storage and adipokines excretion. BAT is enriched in the inner membrane of mitochondria and plays important roles in the non-shivering thermogenesis by uncoupling oxidative phosphorylation from ATP synthesis through UCP1.^{5,6}

In pigs, there is no functional UCP1 protein, resulting in poor thermoregulation and susceptibility to cold stress.⁷ And the temperature maintenance ability of UCP1 knock-in pigs was significantly better than that of wild-type pigs during acute cold exposure. In addition, ectopic UCP1 expression in WAT decreased fat deposition and increase carcass lean percentage.⁸ In rodents, brown fat depots are located in the interscapular region across the lifespan.⁹ This is different from larger adult mammals, including humans, goat and sheep, where brown fat depots are interspersed amongst WAT in the supraclavicular and paraspinous region. In addition, BAT is only abundant during early life, which is located around the internal organs to cope with cold conditions and maintain the core body temperature.¹⁰ After birth, skeletal muscle provides heat through shivering, and thus BAT levels rapidly decline.¹¹ This process is irreversible even if sheep are maintained in a cold environment.¹² However, there is the life-long sensitivity to cold stimulation in rodent intrascapular BAT. Cold exposure of mice activates sympathetic nervous system, which leads to release catecholamines. Then β -adrenergic/p38 MAPK signaling is activated and phosphorylates several transcription factors (ATF2, CREB, and PGC1 α) to trigger the thermogenic program and browning of WAT.¹³ It has been proved in rodents that brown adipocytes are derived from Myf5⁺ mesenchymal precursors using in vivo lineage tracing.¹⁴ And Myf5⁺ progenitor cells into the adipogenic program are determined by the PR domain zinc finger protein 16 (PRDM16), which is essential for the switch between brown adipocytes and myocytes.¹⁵ In addition, bone morphogenetic protein 7 (BMP7) regulates the conversion of Myf5⁺ myogenic progenitors to brown adipocytes by PRDM16.¹⁶ However, the origin of the brown adipocyte in sheep or goats is unclear and remains to be elucidated.

Mechanisms underlying brown and beige fat development have been proved with several pathways including fibroblast growth factor 21 (FGF21), melatonin, and irisin signaling. In rodents, feeding increases the release of FGF21, which promotes energy expenditure and BAT thermogenesis through PPAR α .¹⁷ However, in lamb, feeding promotes heat production, which dependent on

the ambient temperature rather than mediated through BAT thermogenesis.^{18,19} Moreover, feeding in sheep increased thermogenesis in adipose tissue with increased expression of ghrelin.²⁰ Melatonin is a neurohormone, which plays important roles in regulating adipogenesis, WAT browning, and subsequently energy metabolism.²¹ In mice, melatonin reduces weight gain and the total level of cholesterol in serum. And melatonin increases the circulating irisin, which promotes beige adipocyte recruitment and induces the browning of inguinal white adipose tissue (iWAT).²² In sheep, melatonin promotes brown adipocyte differentiation and induced the expression of *UCP1* and *PRDM16*. Moreover, melatonin regulates differentiation of sheep brown adipocytes through activation of AMPK α 1.²³

Many studies have shown that BAT mainly persists in the adrenal fat layer of goats for 1 day and there is a transition from BAT to WAT from 1 day to 1 year.²⁴ When the newborn is exposed to cold from the uterus to the external environment, BAT can effectively regulate the body temperature, and this regulation is considered to be a transition process to adapt to the external environment.¹¹ It has been reported that the ability to regulate body temperature depends on the BAT content of the animal at birth.²⁵ Furthermore, the survival rate of lambs is closely related to their ability of no shivering thermogenesis in BAT.²⁶ It has been reported previously that cold exposure induces browning of iWAT in mice, which alters the lipidomic and transcriptomic profiles involved in thermogenesis, fatty acid elongation, and fatty acid metabolism pathways.²⁷ Held et al indicated that aging affects the rhythmicity of lipid classes including glycerolipids, glycerophospholipids, sterol lipids, and sphingolipids in WAT and BAT of young and old mice.²⁸ In human, Kulterer et al have identified the lipid molecules during cold exposure in both BAT-positive and BAT-negative individuals. They found that the circulating levels of eicosapentaenoic acid, linoleic acid, and 12,13-diHOME are increased only in BAT-positive subjects exposed to cold.²⁹ However, few studies have investigated the changes of transcriptomic and metabolomics between BAT and WAT. In this study, we used RNA sequencing combined with liquid chromatography and mass spectrometry-based metabolomics to determine the changes in BAT metabolomics composition and gene expression. We found that there were significantly changed in the content of fatty acids, amino acids, and carbohydrates in BAT. These changes were accompanied by citric acid cycle (TCA), fatty acid desaturation, and degradation pathways. In addition, L-carnitine plays a critical role in the regulation of brown adipocyte differentiation and thermogenesis through activating AMPK pathway.

2 | MATERIALS AND METHODS

2.1 | Ethics statement

All research involving animals was conducted according to the regulation proposed by the Institutional Animal Care and Use Committee at Sichuan Agricultural University, under permit No. DKY-2018202043.

2.2 | Animals

The Chuanzhong black goats were raised at the breeding center of the Sichuan Agricultural University, Ya'an, China (~1000 m altitude, 103.00°E, 29.98°N), annual temperature is 14°C in average whereas humidity is 52%. The goats were reared according to feeding standard (forage to concentrate ratio, 70:30) twice per day at 07:00–09:00 and 16:00–18:00 and water ad libitum. In this study, Chuanzhong black goats were not related to each other. A total of 16 female goats were used in two stages (1 day and 1 year after birth, 8 individuals at each stage, denoted as D1 and Y1, respectively). All goats were fasted overnight and then injected intramuscularly by using 0.1 ml/kg BW su mian xin (xylazine hydrochloride, Shengda, Changchun, China). Under complete anesthesia, all goats were sacrificed by arterial bleeding. After slaughtering, perirenal adipose tissues were dissected, chilled with liquid nitrogen immediately, and frozen at –80°C.

2.3 | Transcriptomic profile analysis

To provide a comprehensive transcriptome information between BAT and WAT, we selected perirenal adipose tissue samples from four goats at each stage. Total RNA was extracted from perirenal adipose tissue using TRIzol (Invitrogen, CA, USA) and their concentration, purity and integrity were then determined as preparations for RNA-seq. Then, eight sequencing libraries were constructed by NEBNextR UltraTM RNA Library Prep Kit (NEB, MA, USA) according to the instructions and sequenced on an Illumina HiSeq 4000 platform (Illumina, CA, USA). All the transcriptome data have been deposited in the Sequence Read Archive (SRA) database (Accession no. PRJNA547456).

After clean reads were harvested through removing low quality (containing an adapter and had over 10% of poly-N[undefined base]) reads, HISAT2³⁰ was used to align clean reads to goat reference genome (SAR).³¹ The String Tie (Version 1.3.1) was used to assemble the mapped reads from each library.³² In addition, gene expression level was calculated using FPKM (Fragments Per Kilobase of transcript per Million fragments mapped) and

Cuffdiff (v2.1.1)³² software was used to calculate FPKM value. Differentially expressed genes profiles between two groups were identified using the DESeq R package³³ with false discovery rate (FDR) < 0.05 and fold change (FC) > 2. Functional enrichment analysis was performed through the GO term enrichment database and KEGG pathway.

2.4 | Metabolomics analysis

Initially, samples were extracted with extraction solution (40:40:20, v/v/v, methanol:acetonitrile:water), and then were transferred into LC/MS glass vial separately for metabolomic analysis. The UHPLC system (Agilent Technologies, USA) was used for LC-MS/MS analysis equipped with a UPLC BEH Amide column (2.1 × 100 mm, 1.7 μm) (Waters, USA) and a Triple TOF 6600 mass spectrometer. Eluent A consisted of 25 mM NH₄Ac and 25 mM NH₄OH. Eluent B was of acetonitrile and then carried as follows: 95% B at 0 min, 95% B at 0.5 min, 65% B at 7 min, 40% B at 8 min, 40% B at 9 min, 95% B at 9.1 min, and 95% B at 12 min. The flow rate was 0.5 ml/min⁻¹. The conditions of mass spectrometry were set as following: ion source gas 1 (60 Psi), ion source gas 2 (60 Psi), curtain gas (35 Psi), source temperature of 600°C, and capillary voltage of 5000 V or –4000 V in positive or negative modes, respectively.

For metabolites identification and annotation, R package CAMERA (version 1.42.0)³⁴ was used for peak annotation. Then, metabolites data were analyzed by the principal component analysis (PCA) and orthogonal partial least-squares-discriminant analysis (OPLS-DA) using R package ropls (version 1.18.0). Differences in the metabolites between WAT and BAT were determined using Student's *t* test or Mann-Whitney *U* test. First, we analyzed the normal distribution of data by the Kolmogorov-Smirnov test. If the data fitted to normal distribution, statistical significance was determined by Student's *t* test. If the data did not fit to normal distribution, Mann-Whitney *U* test was used between two groups. The fold change of each metabolite was calculated by comparing the mean value of the peak area obtained from BAT samples to that from the WAT samples. *p* < .05, Fold change > 2, and VIP > 1 were set as the threshold for significantly changed metabolites.

2.5 | Brown preadipocytes culture and L-carnitine treatment

Brown preadipocytes were isolated from perirenal adipose tissues of Chuanzhong black goats (1 day after birth) using 1% type I-collagen enzyme digestion for 1 h at 37°C. And white preadipocytes were isolated from perirenal adipose

tissues of Chuanzhong black goats (1 year after birth) using 1% type I-collagen enzyme digestion for 1.5 h at 37°C. Then, preadipocytes were cultured in DMEM medium with 10% FBS, 200 U/ml penicillin, and 200 µg/ml streptomycin. Preadipocytes were induced differentiation in DMEM containing insulin (100 nM), T3 (1 nM), dexamethasone (1 mM), and IBMX (250 mM) for four days. Next, the cells were maintained in DMEM medium containing insulin and T3 for four days. L-carnitine (Sigma-Aldrich, MO, USA) was added into medium throughout the differentiation process and supplemented with 0, 200, 400, and 600 µM, respectively. To determine whether L-carnitine promotes brown adipose thermogenesis by activating AMPK, cells were treated with L-carnitine (600 µM) in the presence or absence of 2 µM Compound C (Selleck, TX, USA) throughout the differentiation stages. Compound C is an inhibitor of AMPK, which is widely used to inhibit AMPK activity to study the function of physiologically active AMPK in vitro and in vivo.³⁵

2.6 | siRNA interference and transfections

Three small interfering RNA fragments (siRNA) against AMPK α and negative control were synthesized by Ribobio (Guangzhou, China). AMPK α siRNA oligonucleotides: S1, GAAGAGAAATTCAGAACCT; S2, CACATG AATGCAAAGATAG; S3, AGAAGTAATTCAGGAAGA. The transfection was conducted siRNA by LipofectamineTM 3000 Reagent (Invitrogen, USA). After 48 h transfection, cells were kept in the presence or absence of L-carnitine (600 µM) differentiation medium containing insulin (100 nM), T3 (1 nM), dexamethasone (1 mM), and IBMX (250 mM) for four days.

2.7 | Oil red O staining and triglyceride content assay

Adipocytes were fixed with 4% formaldehyde for 30 min. Then, ORO working solution was added to each well for 30 min. Finally, the adipocytes were washed twice in water and then photographed. In addition, 100% isopropanol was used to extract Oil Red O dye from adipocytes at room temperature. The OD values were measured at 490 nm on a microplate reader (Thermo, America).

2.8 | Mitochondrial DNA content

For mitochondrial DNA content analysis, qPCR analysis was performed by the total DNA extracted from goat

brown adipocytes. Mitochondrial DNA content was calculated by the ratio of amplification between mitochondrial DNA (ATP6 and COX1) and nuclear DNA (GCG).

2.9 | qPCR

1 µg of total RNA was reverse-transcribed by Prime ScriptTM reagent Kit with gDNA Eraser (Takara, Tokyo, Japan). Then, SYBR Premix Ex TaqTM II (TaKaRa, Otsu, Japan) was used for qPCR in a Bio-Rad CFX96 system (Bio-Rad, Hercules, USA). The conditions of circulation were formed by a single cycle for 3 min at 95°C, then followed by 40 cycles of 20 s at 94°C, 20 s at 54–61°C, 15 s at 72°C, and final extension for 5 min. Primer pairs were designed using Primer Premier 6.0 software based on the goat-related genes and primer sequences was shown in Table S1. The specificity of PCR products was confirmed by melting curve analysis. The PCR fragments were purified and sequenced to confirm the correct amplification of target genes. Our previous study has shown that TBP is recommended as one of the most suitable reference genes in qPCR during the transition process from BAT to WAT.³⁶ In this study, the comparative Ct ($2^{-\Delta\Delta Ct}$) method was used to calculate gene expression and TBP gene was used as an internal control.

2.10 | Western blotting

Protein samples were separated in 10% SDS-PAGE, transferred to PVDF membranes, and blocked in 5% nonfat milk for 2 h, then incubated with primary antibody (rabbit anti-UCP1, 1:1000 dilution, Abcam, CA, USA; mouse anti-AMPK alpha, proteintech, MI, USA; mouse anti-Phospho-AMPK alpha (Thr172), Affinity, Changzhou, China) overnight at 4°C. The cells were incubated at room temperature for 2 h with the secondary antibody (HRP-labeled goat anti-rabbit IgG, 1:1000 dilution, Beyotime, Shanghai, China). Finally, an ECL detection system (Beyotime, Shanghai, China) was used to detect immunoreactive proteins.

3 | RESULTS

3.1 | Characterization of the goat WAT and BAT perirenal fat at two stages

To verify the adipose tissue types of perirenal fat in two stages, we characterized the perirenal fat at two stages by hematoxylin-eosin (H&E) staining analysis. As shown in Figure 1A, multilocular cells in the perirenal region were detectable in 1-day goats and unilocular/

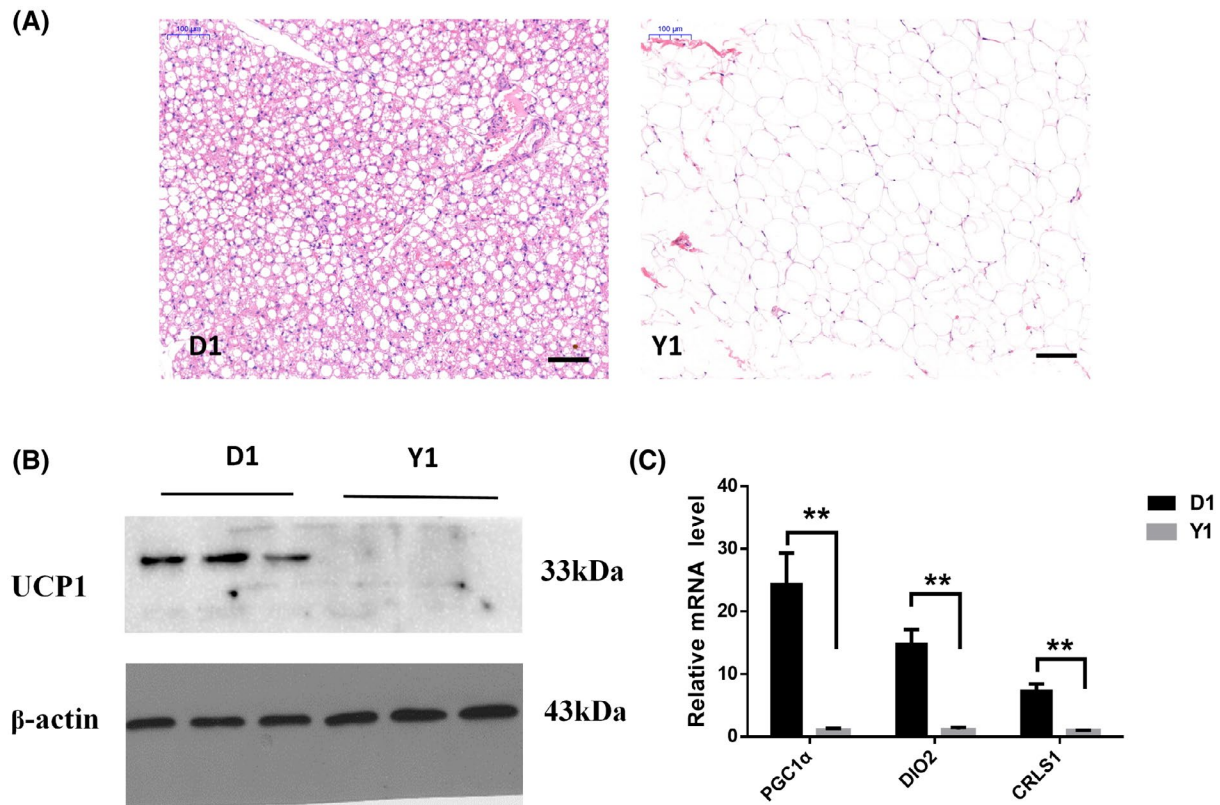


FIGURE 1 Characterization of goat perirenal fat at 1 day (D1) and 1 year (Y1) after birth. (A) Hematoxylin-eosin staining analysis of perirenal fat at 1 day (D1) and 1 year (Y1) after birth. Scale bars: 100 μ m. (B) The level of uncoupling protein 1 (UCP1) was determined by western blotting. (C) The mRNA expression level of PGC1 α , DIO2, and CRLS1 genes in the perirenal fat from two stages by qPCR. Values are expressed as mean $\pm$ SD of $n = 8$. ** $p < .01$ relative to Y1 level

larger lipid morphology were found in 1-year goats. Furthermore, we detected the protein level of UCP1, which is the BAT key thermogenic factor. The immunoblotting results indicated that strong expression of UCP1 protein was only observed in perirenal fat at D1 and undetectable in perirenal fat at Y1 (Figure 1B). We subsequently analyzed the expression of BAT marker genes between two stages. As expected, proliferator-activated receptor- γ coactivator-1 α (PGC1 α), iodothyronine deiodinase 2 (DIO2), and cardiolipin synthase 1 (CRLS1) genes were significantly highly expressed at D1. (Figure 1C). These results suggest that the perirenal fat in 1-day goats was BAT, while the perirenal fat in 1-year goats was WAT.

3.2 | Dramatically changes the expression of genes involved in fatty acid metabolism between BAT and WAT

Based on a false discovery rate (FDR) < 0.05 , and fold change (FC) > 2 , totally 3557 differentially expressed genes were found, of which 2072 and 1485 genes show down- and up-regulated expression patterns in BAT compared to WAT,

respectively (Figure 2A). Notably, several genes reported previously as thermogenesis-related genes were significantly upregulated in BAT, such as UCP1, PGC1 α , PGC1 β , DIO1, DIO2, carnitine palmitoyltransferase 1A (CPT1A), carnitine palmitoyltransferase 1B (CPT1B), and Glycerol-3-Phosphate Acyltransferase 4 (GPAT4) (Figure 2B). KEGG enrichment analyses of up-regulated genes in BAT revealed a significant enrichment of several pathways, including TCA cycle, fatty acid metabolism, fatty acid degradation, PPAR signaling pathway, valine, leucine and isoleucine degradation, and glycolysis/gluconeogenesis (Figure 2C).

3.3 | Metabonomic analysis revealed the different metabolites profiles of BAT and WAT

To explore the substantial differences between two types of adipose tissue, we analyzed the metabolites profiles of BAT and WAT. As shown in Figure 3A, the principal component analysis could distinguish clusters of WAT and BAT group. PC1 and PC2 could explain 76.4% and 12.8% of the variation. Furthermore, the OPLS-DA plot shows a clear separation between WAT and BAT

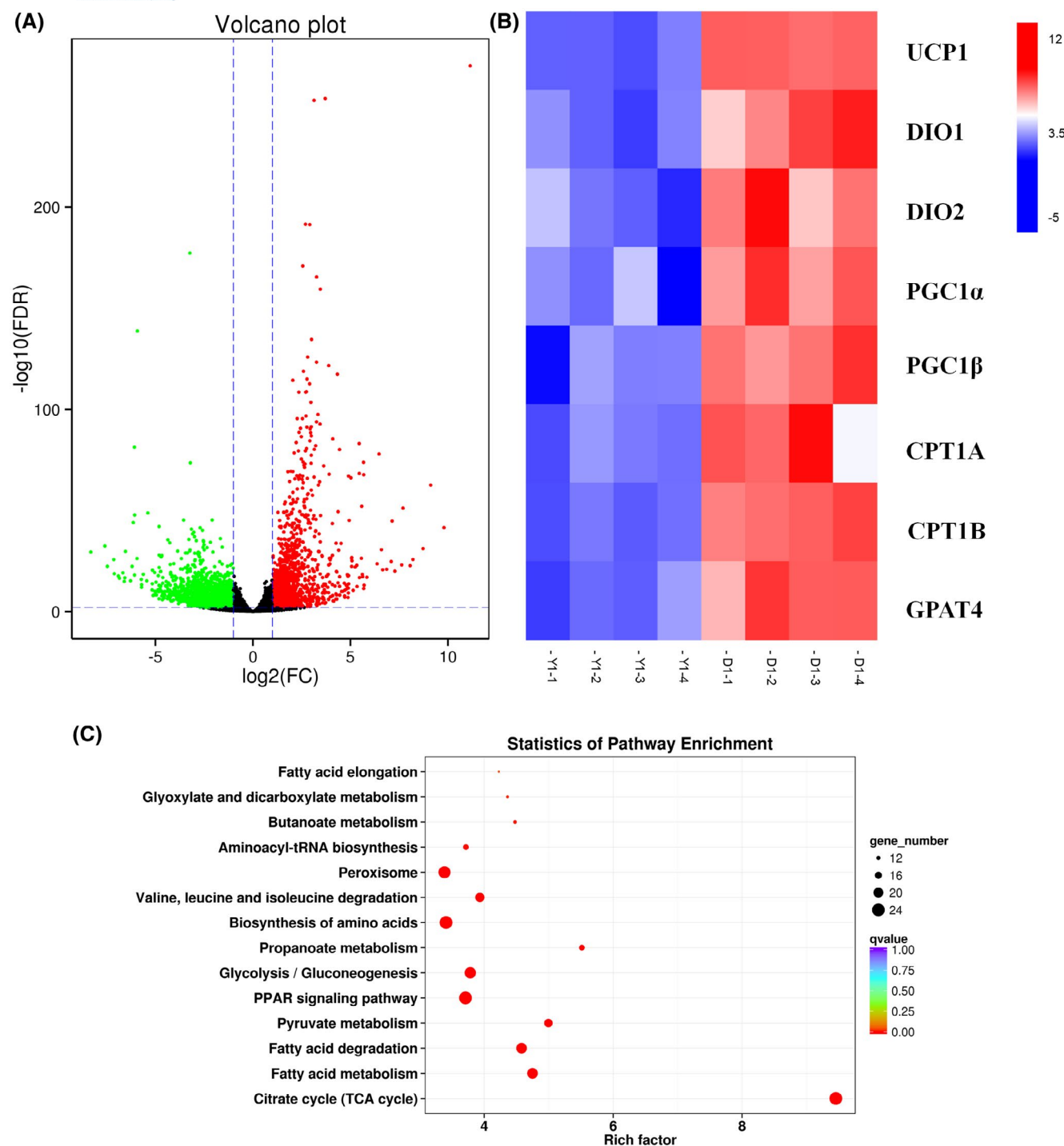


FIGURE 2 Volcano plot and KEGG pathway analysis of differential expression genes between white adipose tissue and brown adipose tissue (BAT). (A) Volcano plot of differential expression genes among two stages in the pairwise comparisons. The green, red, black dots represent the down-regulated, up-regulated and the unchanged genes respectively. (B) Heatmap of BAT thermogenic genes from the RNA-seq data. (C), Statistics of pathway enrichment analysis of differential expression genes. Each point represents a KEGG pathway. The X-axis is the enrichment factor, representing the ratio of the differential expression genes and all genes that annotated to the pathway

($R^2X = 0.677$, $R^2Y = 0.997$, $Q^2 = 0.974$) (Figure 3B). To further determine the relationships between BAT and WAT samples, all of the significantly changed metabolites were performed a clustering analysis by heatmap. The samples from BAT and WAT were divided into two

clusters and the eight replicates for each stage clustered together (Figure 3C). A total of 157 metabolites were significantly changed between WAT and BAT, including 81 upregulated and 76 downregulated metabolites (Table S2). Further analysis showed that the metabolites

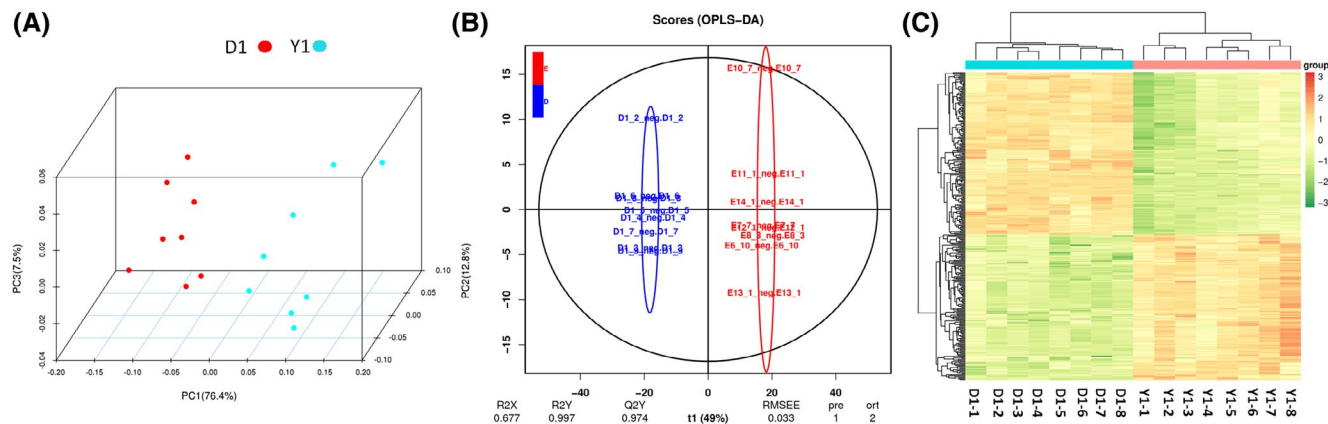


FIGURE 3 Multivariate data analysis for metabolites in white adipose tissue (WAT) and brown adipose tissue (BAT). (A) Principal component analysis of metabolites in WAT and BAT. (B) Orthogonal partial least-squares-discriminant analysis of metabolites in WAT and BAT. Blue and red symbols represent BAT and WAT samples, respectively. (C) Heatmap of the significantly altered metabolites between BAT and WAT samples

TABLE 1 List of selected differently expressed metabolites in fatty acid degradation and desaturation between goat brown adipose tissue (BAT) and white adipose tissue (WAT) (BAT vs. WAT)

Metabolites name	Fold change	<i>p</i> value	VIP value ^a	Regulated ^b
Eicosadienoic acid (C20:2)	5.1147	6.52 E-07	1.3916	Up
γ-Linolenic acid (C20:3)	4.1614	1.35 E-06	1.3298	Up
Arachidonic acid (C20:4)	2.7233	1.66 E-06	1.3524	Up
Adrenic acid (C22:4)	6.7316	1.04 E-05	1.3423	Up
Docosahexaenoic acid (C22:6)	2.1601	3.64 E-05	1.2499	Up
L-carnitine	19.9019	1.32 E-04	1.3288	Up
L-Palmitoylcarnitine	13.0001	5.64 E-04	1.2809	Up

^aVIP value, variable influence on projection (VIP) values were obtained from the orthogonal partial least-squares-discriminant analysis (OPLS-DA).

^b*p* < .05 and VIP > 1 were set as the threshold for significantly changed metabolites.

consisted of fatty acids, amino acids, carbohydrates, and organic chemicals. Interestingly, five polyunsaturated fatty acids (Eicosadienoic Acid, C20:2; γ-Linolenic acid, C20:3; Arachidonic Acid, C20:4; Adrenic acid, C22:4; and Docosahexaenoic acid, C22:6) were found to be abundant in BAT (Table 1). In addition, we found that some fatty acylcarnitines including L-carnitine and L-palmitoyl-carnitine were upregulated in BAT (Table S3).

3.4 | Fatty acid desaturation and degradation pathways are induced in BAT

Our RNA-seq and metabolomic results revealed a significant enrichment of fatty acid desaturation and degradation pathways. When analyzing the individual fatty-acyl-chain composition in two types of adipose tissues, we found that palmitic acid (C16:0), oleic acid (C18:1), linoleic acid (C18:2), α-linolenic acid (C18:3),

Stearidonic acid (C18:4), arachidic acid (C20:0), and behenic acid (C22:0) were more abundant in WAT. In addition, the *PPT1* and *FASN* genes, which involved in fatty acid chain extension, were also upregulated in WAT (Figure 4A). Interestingly, we can observe L-carnitine and L-palmitoyl-carnitine, which are intermediates in the fatty acid degradation pathway, were significantly upregulated in BAT. In addition, we found that the expression of related genes involved in fatty acid degradation, for example, *ACSL3*, *ACSL5*, *ACOX1*, *HADHA*, *HADHB*, *CPT1A*, and *CPT1B*, were up-regulated in BAT (Figure 4B). Notably, significant increases in the concentration of Eicosadienoic Acid (C20:2), γ-Linolenic acid (C20:3), Arachidonic Acid (C20:4), Adrenic acid (C22:4), and Docosahexaenoic acid (C22:6) in BAT. The expression of *ELOVL3* and *ELOVL5* genes was upregulated in BAT, and the two genes encoding the enzymes exhibits activity toward saturated and unsaturated C18-C22 acyl-CoA substrates (Figure 4C).

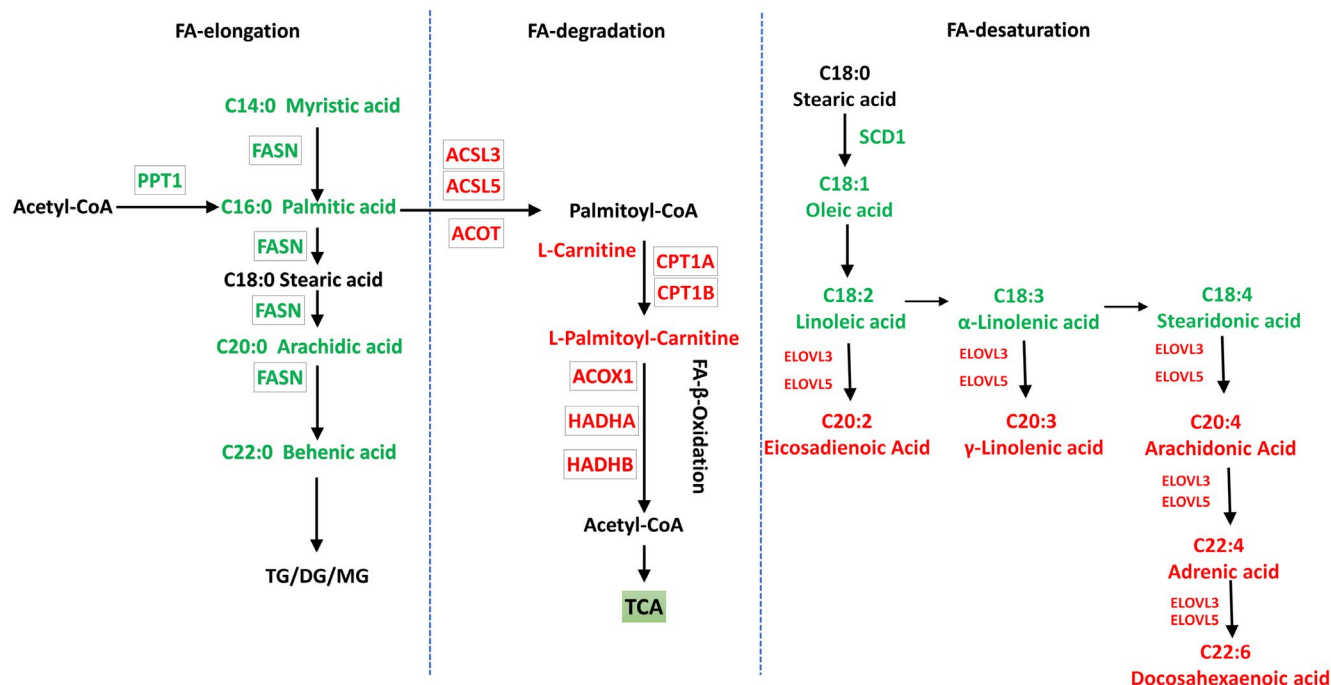


FIGURE 4 The fatty acid elongation, degradation, and desaturation pathways are induced in brown adipose tissue (BAT). Selected fatty acid elongation, degradation, and desaturation metabolic reactions, with indications of genes and metabolites significantly differential expressed between BAT and white adipose tissue. Colors indicate down-regulated (green) or up-regulated (red) expression of genes and metabolites in BAT

3.5 | L-carnitine promotes goat brown adipocyte differentiation and thermogenesis

To further demonstrate the effect of L-carnitine on goat brown adipocyte differentiation and thermogenesis, primary brown preadipocytes were incubated in the presence or absence of L-carnitine at a concentration of 0, 200, 400, and 600 μ M. At day 8 of differentiation, Oil red O staining showed that L-carnitine increased lipid accumulation in a dose-dependent manner with increasing concentrations of L-carnitine from 200 to 600 μ M (Figure 5A,B). The mRNA expression of *FABP4*, *PPARG*, and *FASN* was markedly ($p < 0.05$) induced by L-carnitine treatment at concentration of 400 and 600 μ M (Figure 5C). Furthermore, at day 8 of differentiation, L-carnitine significantly ($p < .01$) up-regulated the expression levels of brown adipose marker genes such as *UCP1*, *PGC1 α* , *DIO2*, and *CRLS1* (Figure 5D). And L-carnitine significantly ($p < .01$) increased mitochondrial related genes, including *COX1* and *ATP6* (Figure 5E). These results demonstrate that L-carnitine plays important roles in goat brown adipogenesis and thermogenesis.

3.6 | L-carnitine decreased lipid accumulation through inducing lipolysis and thermogenesis in white adipocytes

We next examined the role of L-carnitine on white adipocytes differentiation and thermogenesis. Interestingly, L-carnitine decreased lipid accumulation with 600 μ M L-carnitine (Figures 6A,B). However, L-carnitine failed to decrease the expression of lipogenic genes (Figure 6C). Furthermore, L-carnitine increased the expression of thermogenic genes (*UCP1*, *PGC1 α* , *DIO2*, and *CRLS1*) (Figure 6D) and lipolysis related genes (*ATGL* and *LPL*) in white adipocytes (Figure 6E). These results suggest that the reduced lipid content by L-carnitine is due to higher lipolysis and thermogenesis activity in white adipocytes.

3.7 | L-carnitine regulates the brown adipose thermogenesis through activating the AMPK signaling pathway

To investigate whether L-carnitine regulated AMPK activation, we determined the levels of total AMPK protein

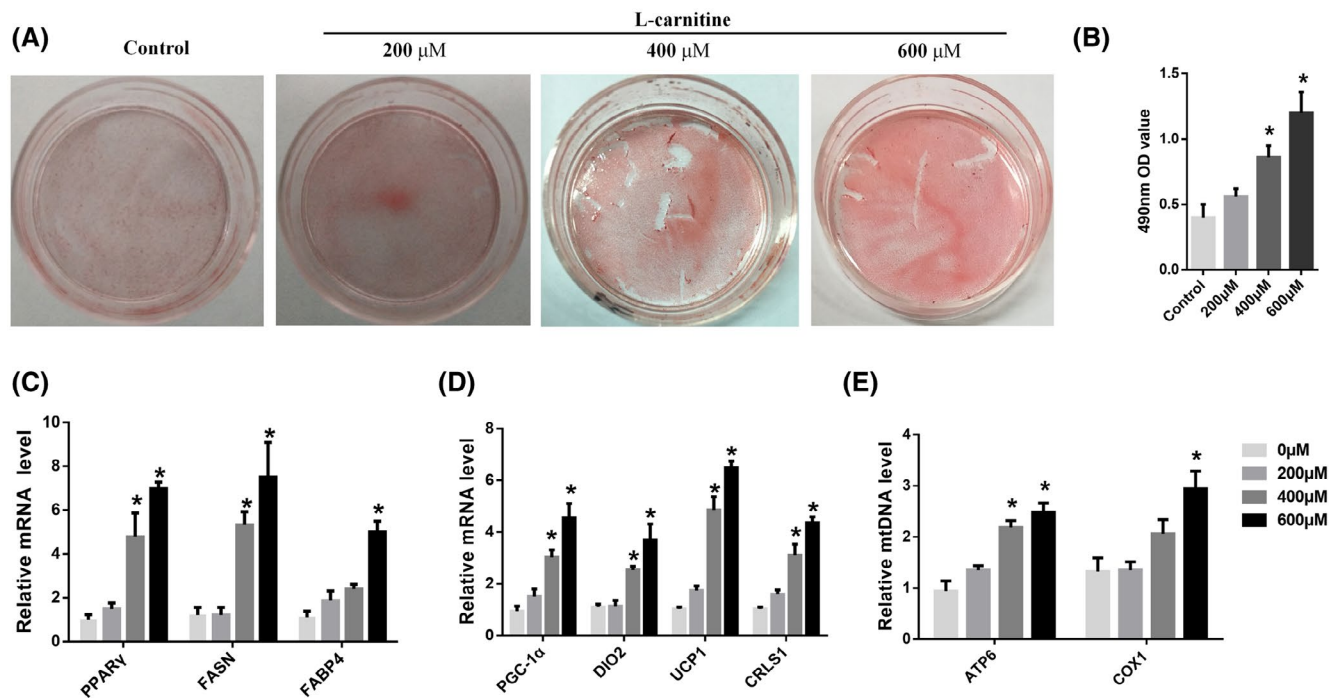


FIGURE 5 Effect of L-carnitine on goat brown adipocyte differentiation and thermogenesis. (A) Oil red O staining, (B) lipid content, and (C) adipogenic genes expression in brown adipocytes treated with L-carnitine (0, 200, 400, and 600 μ M) on day 8 of differentiation. (D) Expression levels of thermogenesis-regulated genes and (E) mitochondrial DNA content in goat brown adipocytes with L-carnitine treatment (0, 200, 400, and 600 μ M) on day 8 of differentiation. Values are expressed as mean $\pm$ SEM. of $n = 6$. ** $p < .01$; * $p < .05$ relative to 0 μ M level

and pAMPK α (T172) in goat brown adipocytes by western blot. The level of pAMPK α (T172) was significantly ($p < .01$) upregulated after L-carnitine treatment from 200 to 600 μ M (Figure 7A,B). There was no obvious difference of total AMPK protein after L-carnitine treatment. In addition, the key thermogenic marker, UCP1 protein, was upregulated after L-carnitine treatment (Figure 7A,C).

An inhibitor of AMPK by Compound C was used to further determine the role of L-carnitine in brown adipose thermogenesis through the AMPK signaling pathway. L-carnitine treatment significantly promoted the phosphorylation of pAMPK α (T172) and the AMPK inhibition by Compound C attenuated the ability of L-carnitine to induce phosphorylation of pAMPK α (T172) (Figure 8A,B). Additionally, Compound C greatly ($p < .01$) repressed L-carnitine-induced the levels of UCP1 protein (Figure 8C) and the mitochondrial DNA content (Figure 8E). The induction of thermogenesis-regulated genes (*UCP1*, *PGC1 α* , *DIO2*, and *CRLS1*) expression by L-carnitine was robustly blunted after L-carnitine and Compound C co-incubation (Figure 8D).

Furthermore, the siRNA-mediated specific knockdown of the AMPK α was performed. We first examined the efficiency of siRNAs. As shown in Figure 9A, the protein levels of AMPK α in the siRNA1 and siRNA2 groups were significantly decreased compared with the negative control group

(NC), and then siRNA1 (S1) was used for subsequent experiments. The results showed that AMPK α knockdown by RNAi attenuated the ability of L-carnitine to induce UCP1 protein (Figure 9B,C). In addition, AMPK α knockdown blunted the ability of L-carnitine to induce the expression of thermogenesis-regulated genes (*UCP1*, *PGC1 α* , *DIO2*, and *CRLS1*) (Figure 9D) and the mitochondrial DNA content (Figure 9E). These results indicated that the existence of AMPK α is necessary for L-carnitine-promoted BAT thermogenesis.

4 | DISCUSSION

Transcriptomics serves as a powerful study method, which could provide many important information through identifying genetic regulatory networks.³⁷ Untargeted metabolomics, which is capable of providing a global insight into small molecules changes in samples and also reflect the functional status of an organism.³⁸ Untargeted metabolomics has a broad coverage on metabolite measurement without a priori knowledge of the constituents. In addition, the advantage of untargeted metabolomics is the discovery of novel biomarkers, thereby providing further insight into underlying physiological mechanisms.³⁹ In this study, we provided a comprehensive metabolomics and transcriptional

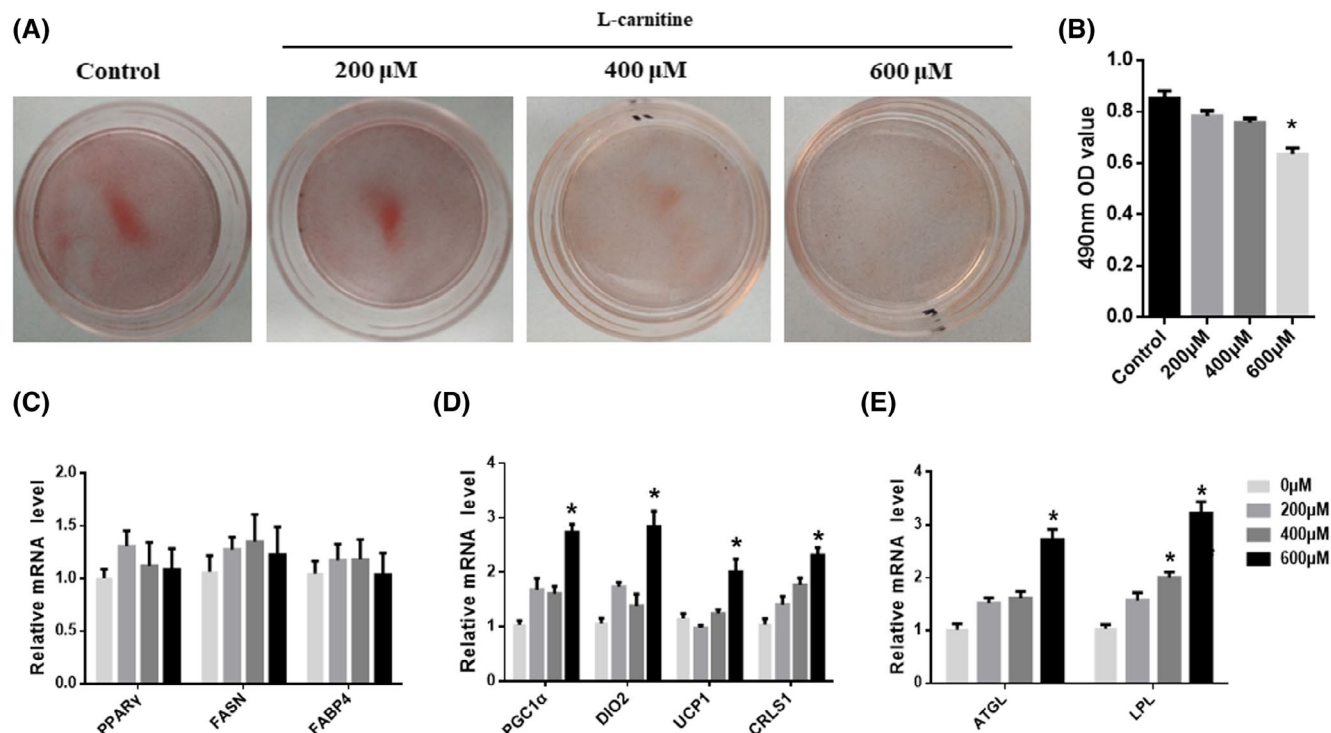


FIGURE 6 L-carnitine decreased lipid accumulation through inducing lipolysis and thermogenesis in white adipocytes. (A) Oil red O staining, (B) lipid content, and (C) adipogenic genes expression in white adipocytes treated with L-carnitine on day 8 of differentiation. (D) Expression levels of thermogenesis-regulated genes and (E) lipolysis related genes in white adipocytes with L-carnitine treatment on day 8 of differentiation. Values are expressed as mean $\pm$ SEM. of $n = 6$. ** $p < .01$; * $p < .05$ relative to 0 μ M level

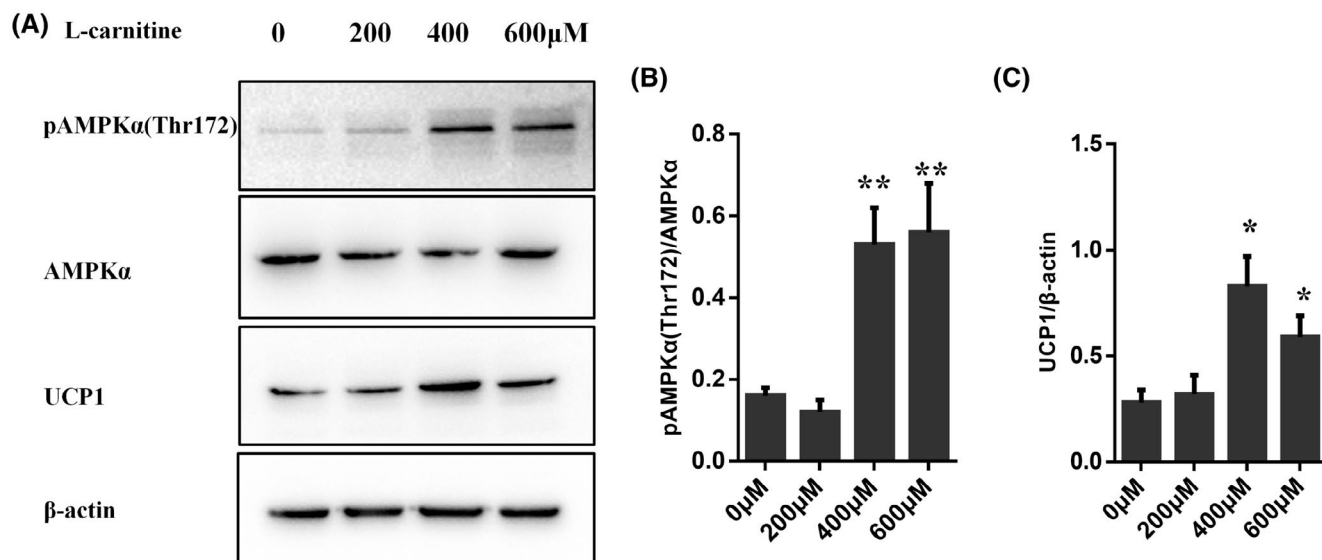


FIGURE 7 The levels of pAMPKα (T172) and UCP1 was induced by L-carnitine in brown adipocytes. (A) Representative western blots and (B, C) quantified results of pAMPKα (T172) and UCP1 in brown adipocytes treated with L-carnitine (0, 200, 400, and 600 μ M) on day 8 of differentiation. Values are expressed as mean $\pm$ SEM. of $n = 6$. ** $p < .01$; * $p < .05$ relative to 0 μ M level

signatures of BAT and WAT through integrating the two “omics” data. For a more detailed pathway analysis of different expressional genes and metabolites, we found that the fatty acid elongation, degradation, and desaturation as a strongly regulated pathway between BAT and WAT.

In our study, a wide variety of long chain fatty acids enriched in WAT indicates the strong lipid synthesis capacity of WAT as long-chain fatty acids ($n > 12$) are usually incorporated into triacylglycerol, phospholipids, and cholesterol esters to synthesize complex lipid.^{40,41} Our integrated results

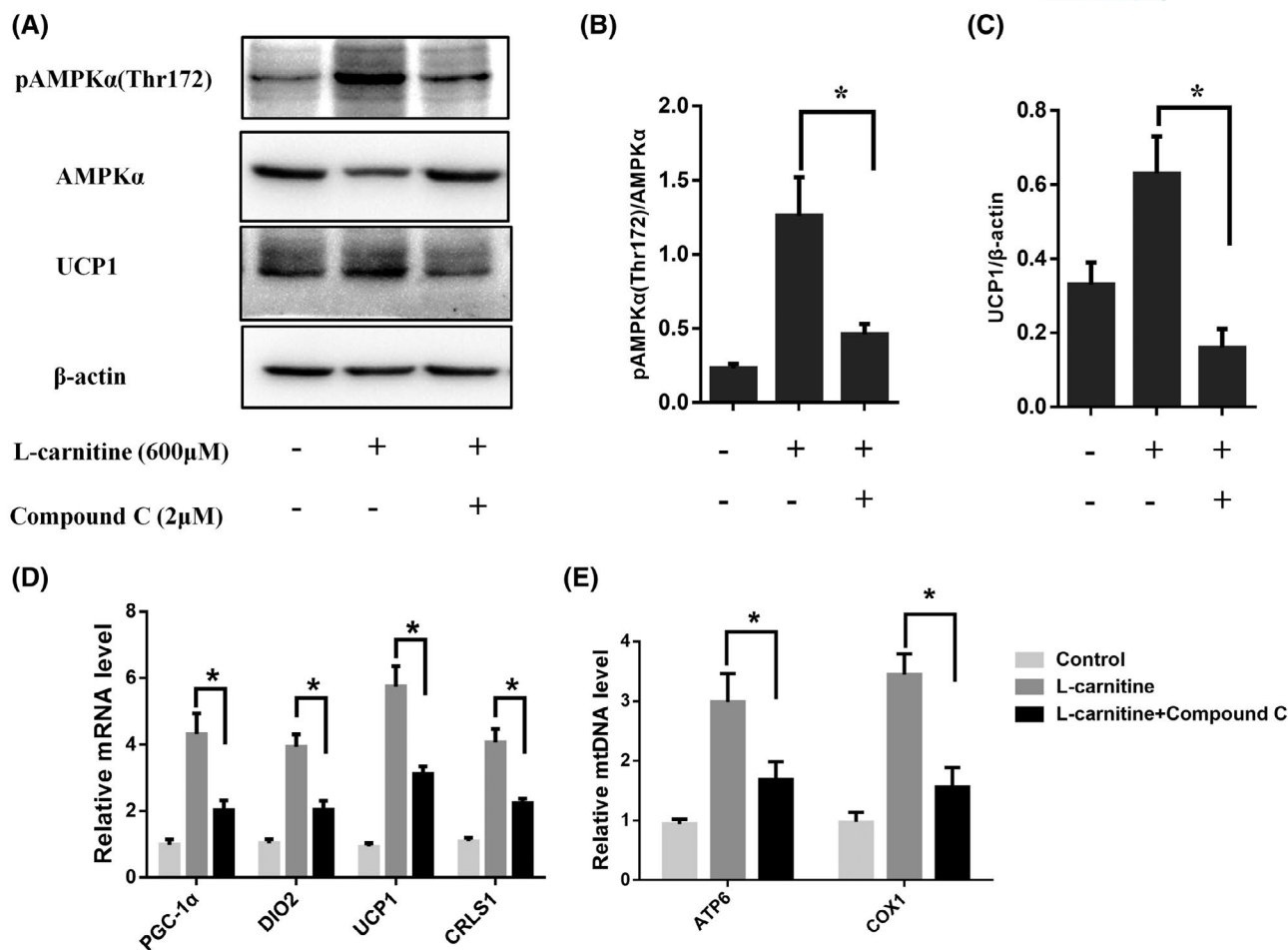


FIGURE 8 L-carnitine promotes the brown adipose thermogenesis through activating the AMPK pathway. (A) Representative western blots and (B, C) quantified results of pAMPKα (T172) and UCP1 in brown adipocytes after L-carnitine/Compound C treatment on day 8 of differentiation. (D) Expression levels of thermogenesis-regulated genes and (E) mitochondrial DNA content. Values are expressed as mean ± SEM. of $n = 6$. ** $p < .01$; * $p < .05$ relative to L-carnitine treatment group

indicate that expression of several genes in fatty acid elongation is induced significantly in WAT. Stearoyl-CoA desaturase (SCD), which located in the endoplasmic reticulum is mainly responsible for the synthesis of monounsaturated fatty acids (oleate and palmitoleate).⁴² Palmitoyl-protein thioesterase, PPT1, is primarily responsible for removing thioester-linked fatty acyl groups that attach to modified cysteine residues in polypeptides, and is responsible for termination of fatty acid chain extension.^{43,44} Fatty acid synthetase, FASN is mainly responsible for catalyzing the biosynthesis of long-chain saturated fatty acids from acetyl-CoA and malonyl-CoA in the presence of NADPH.⁴⁵

Interestingly, we observed a significant increase of eicosadienoic acid (C20:2), γ -linolenic acid (C20:3), arachidonic acid (C20:4), adrenic acid (C22:4), and docosahexaenoic acid (C22:6) in BAT. Fatty acids serve as an important fuel for the adipose-based non-shivering thermogenesis.⁴⁶ Supplementation of Omega-3 fatty acids (EPA and DHA) in HFD feed mouse can reduce lipid accumulation in adipose tissues.⁴⁷ And numerous evidence

suggests some kind of PUFAs, such as, eicosapentaenoic acid (EPA),⁴⁸ docosahexaenoic acid (DHA),⁴⁹ and α -linolenic acid,^{50,51} contribute to promoting thermogenic capacity of WAT and increasing the expression level of *UCP1*. It has been reported previously that cold exposure significant induced fatty acid elongation and desaturation in mouse iWAT.^{27,52} Our results show that expression level of *ELOVL3* and *ELOVL5* genes was upregulated in BAT, suggesting that they are required for enhancing thermogenic potential. In addition, *ELOVL3* and *ELOVL5*, which localized in the endoplasmic reticulum, play important roles in the production of monounsaturated and polyunsaturated long chain fatty acids.⁵³ Future investigations should determine whether the kind of PUFAs contributes to thermogenesis process in BAT.

Increasing evidence has shown that fatty acid oxidation is crucial for the maintenance of BAT thermogenic programming.^{54–56} L-carnitine is an intermediate of fatty acid oxidation, and shown to be responsible for adipose tissue thermogenesis.⁵⁷ In our results, the increased amounts of L-carnitine in BAT also prove its important roles in the regulation of

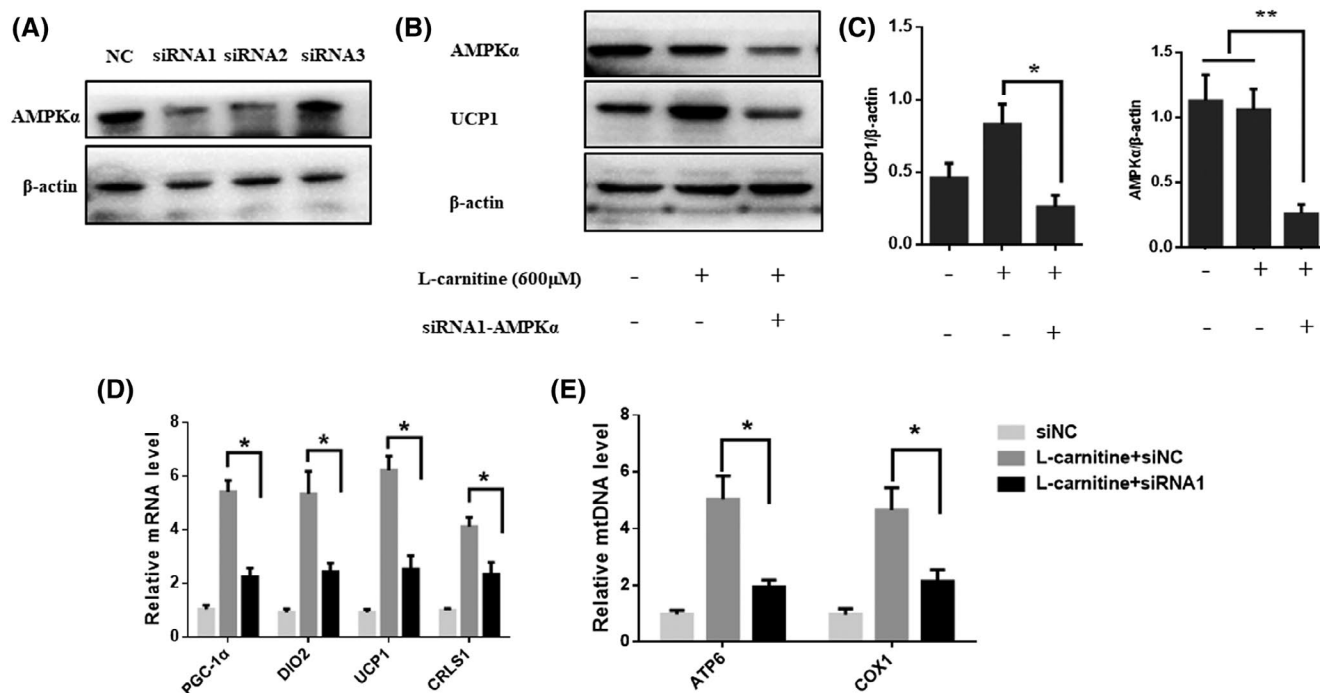


FIGURE 9 AMPK α is necessary for L-carnitine to promote brown adipose thermogenesis. (A) The siRNA-mediated specific knockdown of the AMPK α . (B) Representative western blots and (C) quantified results of AMPK α and UCP1 in brown adipocytes after L-carnitine/siAMPK α . (D) Expression levels of thermogenesis-regulated genes and (E) mitochondrial DNA content. Values are expressed as mean $\pm$ SEM. of $n = 6$. ** $p < .01$; * $p < .05$ relative to L-carnitine treatment group

BAT metabolism. L-carnitine is essential for the transfer of palmitic acid across the inner mitochondrial membrane for subsequent β -oxidation.⁵⁸ L-carnitine administration can improve thermogenic capacity, when mice were challenged with cold.⁵⁹ In the juvenile visceral steatosis (JVS) mouse, carnitine treatment significantly increased the BAT content, body temperature, and the expression of UCP1 mRNA and protein levels.⁵⁷ In this study, we found that L-carnitines have more abundance in the BAT than that in WAT and L-carnitine treatment promotes goat brown adipocyte differentiation and thermogenesis. However, L-carnitine decreased lipid accumulation and induced lipolysis in white adipocytes. Our results suggested that L-carnitine plays an important role on thermogenic function in brown adipocytes. And L-carnitine does not affect adipogenic differentiation but promotes the process of lipolysis and thermogenesis in white adipocytes.

Furthermore, we found that L-carnitine increased phosphorylation of AMPK α . It has been reported that AMPK signaling pathway is involved in regulating the energy homeostasis and thermogenesis of adipose tissue.⁶⁰ AMPK has a reducing effect on body fat accumulation and enhances fatty acid oxidation in liver.⁶¹ And in AMPK $\alpha 1/\alpha 2$ KO mice (AKO), AMPK is essential for mitochondrial thermogenesis while decreasing the expression of thermogenic genes in response to cold stimulation.⁶² AMPK $\alpha 1$ promotes brown adipogenesis by regulating the methylation of PRDM16 promoter⁶³ and enhancing the H2B O-GlcNAcylation of IDH2 gene.⁶⁴ In this study, an

inhibitor of AMPK by Compound C and knockdown of AMPK α by siRNA were used to inhibit activity of AMPK. Then, we found that the actions of L-carnitine on thermogenesis were abolished, indicating that L-carnitine plays an important role in BAT thermogenesis process through the AMPK signaling pathway. These data suggested that dietary L-carnitine regulation may represent a strategy to modulate heat production and energy expenditure in goats. Finally, we recognize that our research has limitation. Although we identified some metabolites are involved in thermogenic regulation of BAT in female goats, the molecular mechanisms of BAT thermogenesis underlying the gender difference still need to be investigated in the future study.

5 | CONCLUSION

In summary, we determined the changes in metabolites between WAT and BAT using omics techniques. We described the significant alterations in the composition of metabolites involved in the fatty acid elongation, degradation, and desaturation. Our study has identified L-carnitine as an intermediate of fatty acid degradation pathway that is necessary for goat brown adipocyte differentiation and thermogenesis. However, L-carnitine reduces lipid content and does not affect adipogenic differentiation but promotes the process of lipolysis and thermogenesis in white

adipocytes. This study provides evidence for the role of metabolites in goat BAT adipogenesis.

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DISCLOSURE

The authors declare no conflicts of interest.

AUTHOR CONTRIBUTIONS

Methodology, conceptualization, and writing-original draft preparation: Y. Wang. Data curation: X. Chen. Validation: W. Fan. Formal analysis: X. Zhang and S. Zhan. Resources: T. Zhong. Software: J. Guo and J. Cao. Resources and investigation: L. Li and H. Zhang. Supervision, funding acquisition, and project administration: L. Wang.

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SUPPORTING INFORMATION

Additional Supporting Information may be found online in the Supporting Information section.

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